Amazigh OUZRIAT

JSPS Postdoctoral Fellow

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LinkedIn · iNSPiRE-HEP Profile

Research Interests

Gravitational waves, compact binary coalescences, cosmology, multi-messenger astronomy, black holes, neutron stars, detector characterization, LISA, Einstein Telescope, Cosmic Explorer.

Research Experience

JSPS Postdoctoral Fellow — **Oct 2026 – Present**
RESCEU, The University of Tokyo — *Tokyo, Japan*
Host: Prof. Kipp Cannon

Ph.D. Research Fellow — **Oct 2022 – Sep 2025**
IP2I Lyon, Claude Bernard University Lyon 1 — *Villeurbanne, France*
Thesis: *Search for Compact Binary Coalescences during the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*
Advisor: Dr. Viola Sordini

Graduate Research Fellow — **Mar 2022 – Jul 2022**
IP2I Lyon, Claude Bernard University Lyon 1 — *Villeurbanne, France*
Project: *Cosmological implications of LIGO-Virgo gravitational wave observations*
Advisor: Dr. Viola Sordini

Graduate Research Fellow — **Apr 2021 – Jul 2021**
CEA Saclay IRFU — *Saclay, France*
Project: *Development of a source of high-energy gamma rays induced by neutron captures for detector calibration*
Advisor: Dr. David Lhuillier

Laboratory Intern — **Sep 2021 – Oct 2021**
CEA Saclay IRFU — *Saclay, France*
Project: *Muon tomography using multiplexed Micromegas detectors and cosmic shear analysis*
Advisor: Dr. Marine Vandebrouck

Undergraduate Research Fellow — **Apr 2019 – Jul 2019**
IP2I Lyon, Claude Bernard University Lyon 1 — *Villeurbanne, France*
Project: *Physical and mathematical modeling of ion-living matter interactions*
Advisor: Prof. Michaël Beuve

Education

Ph.D. in Gravitational-Wave Astronomy — **2022 – 2025**
Claude Bernard University Lyon 1 — *Villeurbanne, France*
Physics & Astrophysics Doctoral School

M.Sc. in Fundamental Physics — **2020 – 2022**
Paris-Saclay University — *Orsay, France*
Track: Nuclei, Particles, Astroparticles & Cosmology (NPAC)

B.Sc. (Licence) in Physics 2017 – 2020
Claude Bernard University Lyon 1 *Villeurbanne, France*

Responsibilities & Collaboration Roles

Science Case Study Team Member 2024 – 2026
LIGO–Virgo–KAGRA (LVK) Collaboration
Responsible for gravitational-wave search analyses for exceptional event papers.

Paper Writing Team Member 2024 – 2026
LVK Collaboration
Contributing to key collaboration papers, including LIGO detector characterization in O4a Open Data Paper and event validation in *GWTC-4: Methods*.

LVK Collaboration Liaison 2023 – 2026
LVK Collaboration
Liaison between Virgo Detector Characterization and Operations / Compact Binary Coalescences (CBC) groups; liaison between MBTA search pipeline and Rates & Populations group.

LVK Observing Run Operations 2023 – 2026
LVK Collaboration
Active member of Rapid Response Team for real-time GW event vetting and Catalog Duty Manager shifts during O4.

Honors & Awards

2026 **JSPS Postdoctoral Fellowship for Research in Japan**
2022 **CNRS/IN2P3 Fellowship "Des 2 Infinis"**
2021 **Best Reporter Award, International Physicists' Tournament (IPT)**
2021 **Institute of the Physics of the Universe Scholarship, Aix-Marseille University *(Declined)***
2021 **First Place, French Physicists' Tournament**
2017 **Regional First Place, Algerian Selection for International Mathematical Olympiad**

Teaching Experience

Teaching Assistant Spring 2024, 2025
Claude Bernard University Lyon 1 *PHY2000E: Fundamentals of Physics*
Physics minor for undergraduate medical students: Optics, electromagnetism, mechanics, and mathematics.

Teaching Assistant Fall 2023, 2024
Claude Bernard University Lyon 1 *PHY2021L: Electromagnetism*
Electrostatics, magnetostatics, electromagnetic induction, and electromagnetic waves for Physics & Chemistry undergraduates.

Teaching Assistant Fall 2023
Claude Bernard University Lyon 1 *TVL1212L: Introduction to Python*
Fundamentals of Python programming: syntax, scientific computing, statistics, and plotting.

Presentations & Seminars

Conference Talk Nov 2025
International Conference on the Physics of the 2 Infinities *Tokyo, Japan*
Unveiling the Sky with Gravitational Waves: Discoveries from GWTC-4

Collaboration Talk **Feb 2025**
Virgo Collaboration Meeting *Cascina, Italy*
Impact of Virgo inclusion on GW sky-localization and effect of glitches

Poster Presentation **Sep 2024**
LVK Collaboration Meeting *Barcelona, Spain*
Search for compact binary coalescences using the offline MBTA pipeline

Conference Talk **Jul 2024**
International Conference: Exploring the Dark Side of the Universe *Noirmoutier, France*
Observations of Compact Binary Coalescences by the LVK Collaboration

Collaboration Talk **Mar 2024**
LVK Collaboration Meeting *Louisiana, USA*
First results of the standard MBTA offline search during O4a

Poster Presentation **Sep 2023**
LVK Collaboration Meeting *Toyama, Japan*
Investigating iDQ integration into the MBTA offline search

Collaboration Talk **Apr 2023**
Virgo Collaboration Meeting *Cascina, Italy*
Investigating iDQ integration into the MBTA offline search

Seminar **Jul 2022**
CEA Institute of Research into the Fundamental Laws of the Universe (IRFU) *Saclay, France*
Open physics problems during the International Physicists' Tournament

Publications

Co-author of **57** publications, including **36** articles in peer-reviewed journals and **52** co-authored papers within the Virgo Collaboration. Complete list available on my iNSPiRE-HEP profile.

References

Dr. Viola Sordini
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